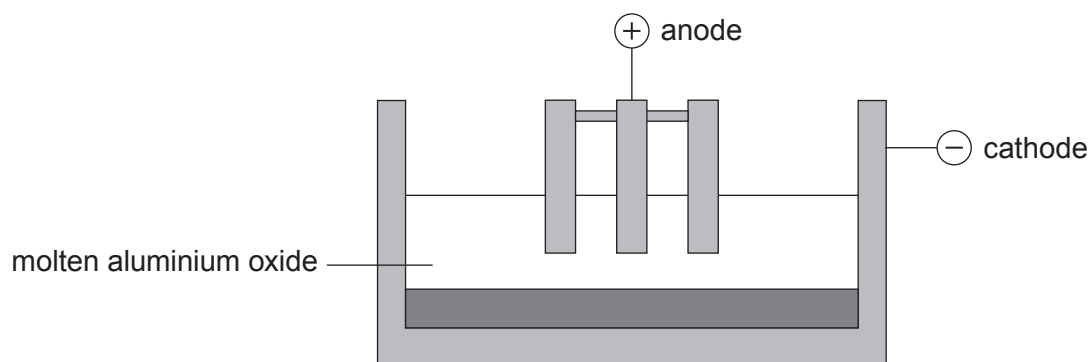


7. Aluminium is extracted from its ore by electrolysis. The diagram shows a cross-section of the cell used for the extraction.



The overall equation for the extraction taking place inside the cell is given below.



Explain the movement of Al^{3+} and O^{2-} ions inside the cell and how the reactions at each electrode demonstrate oxidation and reduction. Include electrode equations in your answer.

[6 QER]

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6

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



28

1

H

Hydrogen

1

7 Li Lithium 3	9 Be Beryllium 4
23 Na Sodium 11	24 Mg Magnesium 12
39 K Potassium 19	40 Ca Calcium 20
86 Rb Rubidium 37	88 Sr Strontium 38
133 Cs Caesium 55	137 Ba Barium 56
223 Fr Francium 87	226 Ra Radium 88

11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86

59 Co Cobalt 27	56 Fe Iron 26	55 Mn Manganese 25	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30
103 Rh Rhodium 45	101 Ru Ruthenium 44	99 Tc Technetium 43	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48
192 Ir Iridium 77	190 Os Osmium 76	186 Re Rhenium 75	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80
181 Ta Tantalum 73	184 W Tungsten 74	181 Re Rhenium 75	190 Os Osmium 76	195 Pt Platinum 78	201 Hg Mercury 80
179 Hf Hafnium 72	179 Ta Tantalum 73	181 Ta Tantalum 73	184 W Tungsten 74	190 Os Osmium 76	195 Pt Platinum 78
139 La Lanthanum 57	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	190 Os Osmium 76	201 Hg Mercury 80
89 Y Yttrium 39	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	101 Ru Ruthenium 44	103 Rh Rhodium 45
45 Sc Scandium 21	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	56 Fe Iron 26

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number